

Lecture Slide – 14
Numerical Integration

NUMERICAL INTEGRATION

Integration of $f(x)$ from a to b is $\int_a^b f(x)dx$

- If we divide $[a, b]$ into n subintervals of equal length
- $\Delta x = (b - a)/n$, we have:

$$\int_a^b f(x)dx \approx \sum_{i=1}^n f(x_i^*)\Delta x$$

- where x_i^* is any point in the i_{th} subinterval $[x_{i-1}, x_i]$.

L_n APPROXIMATION

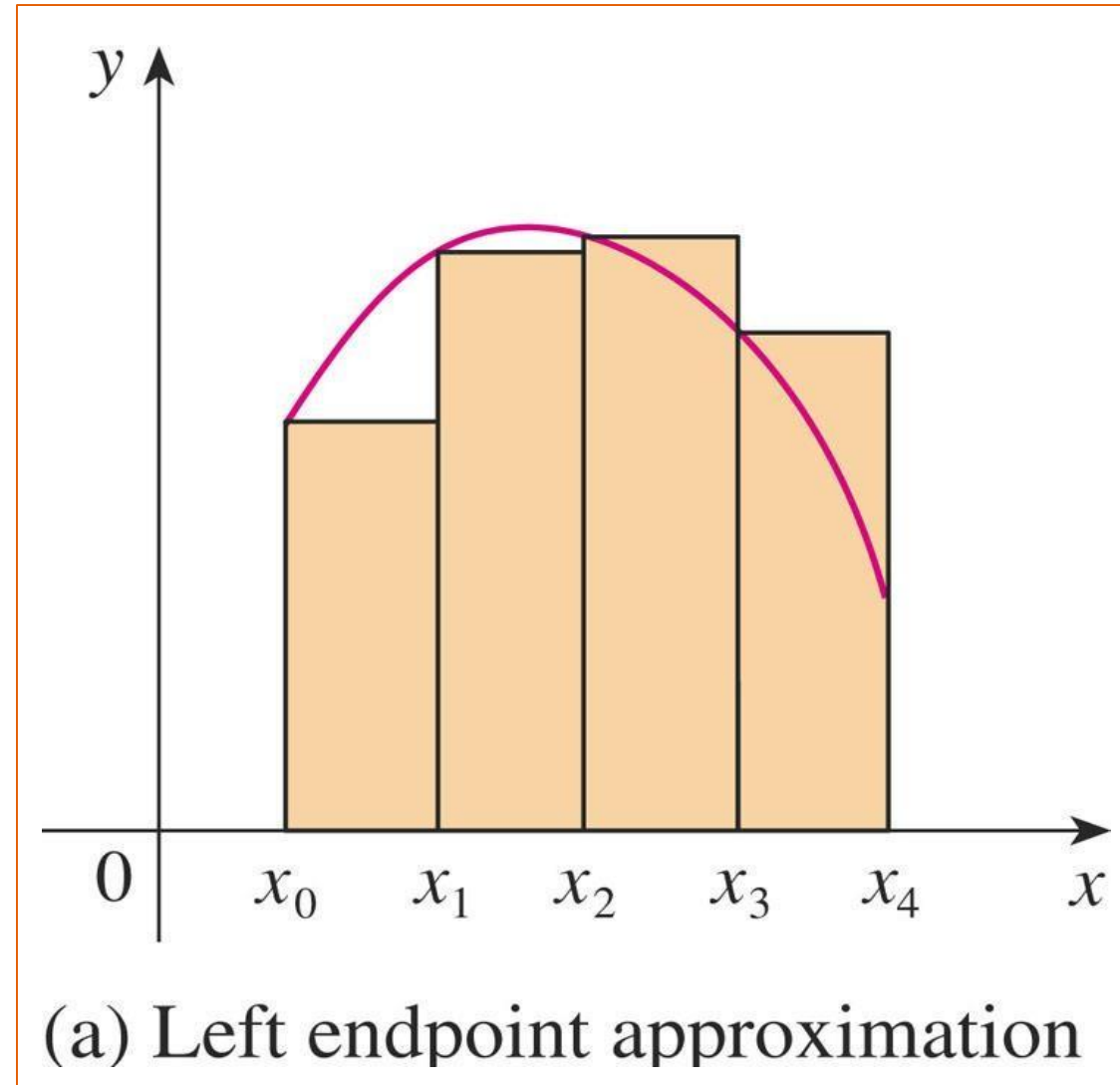
- If x_i^* is chosen to be the left endpoint of the interval, then $x_i^* = x_{i-1}$ and we have:

$$\int_a^b f(x)dx \approx L_n = \sum_{i=1}^n f(x_{i-1})\Delta x \quad \text{Equation-1}$$

- The approximation L_n is called the left endpoint approximation.

L_n APPROXIMATION

- (if $f(x) \geq 0$), $f(x_{i-1}) \times \Delta x$ represents an area of a rectangle, and equation 1, represents the sum of the rectangles, which is an approximation of the area under the curve (shown in Fig (a)).



APPROXIMATE INTEGRATION

Example 1(a)

- ▶ Approximate the integral of $f(x) = 1/x$ with $n = 5$, using left end-point approximation: $\int_1^2 (1/x) dx$
- ▶ With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
- ▶ So, L_n the left endpoint approximation is :

$$\begin{aligned} \int_1^2 \frac{1}{x} dx &\approx L_5 = [f(1) + f(1.2) + f(1.4) + f(1.6) + f(1.8)] \times 0.2 \\ &= 0.2 \left(\frac{1}{1} + \frac{1}{1.2} + \frac{1}{1.4} + \frac{1}{1.6} + \frac{1}{1.8} \right) = 0.745634921 \quad (0.693147) \end{aligned}$$

APPROXIMATE INTEGRATION

- ▶ In Example 1(a), we deliberately chose an integral whose value can be computed explicitly so that we can see how accurate the left end-point approximation is.
- ▶ By the Fundamental Theorem Calculus,

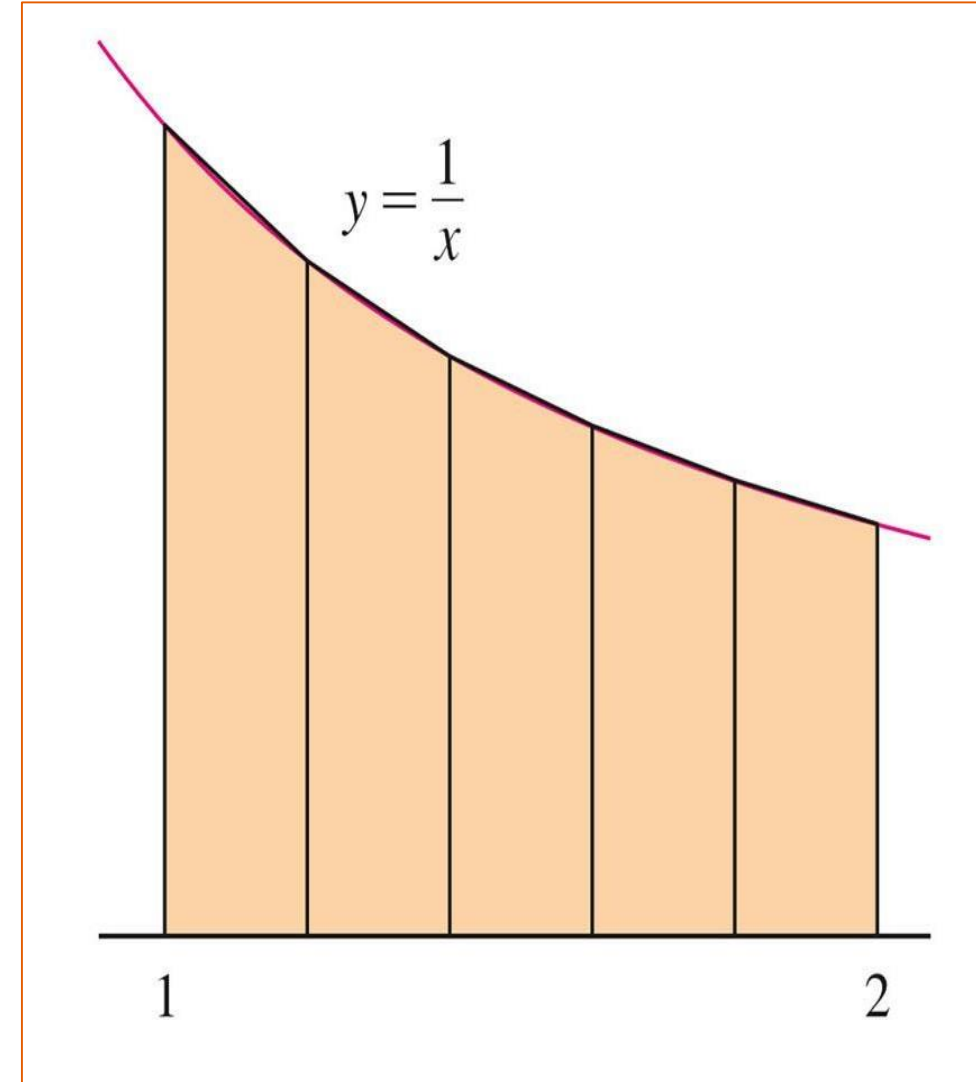
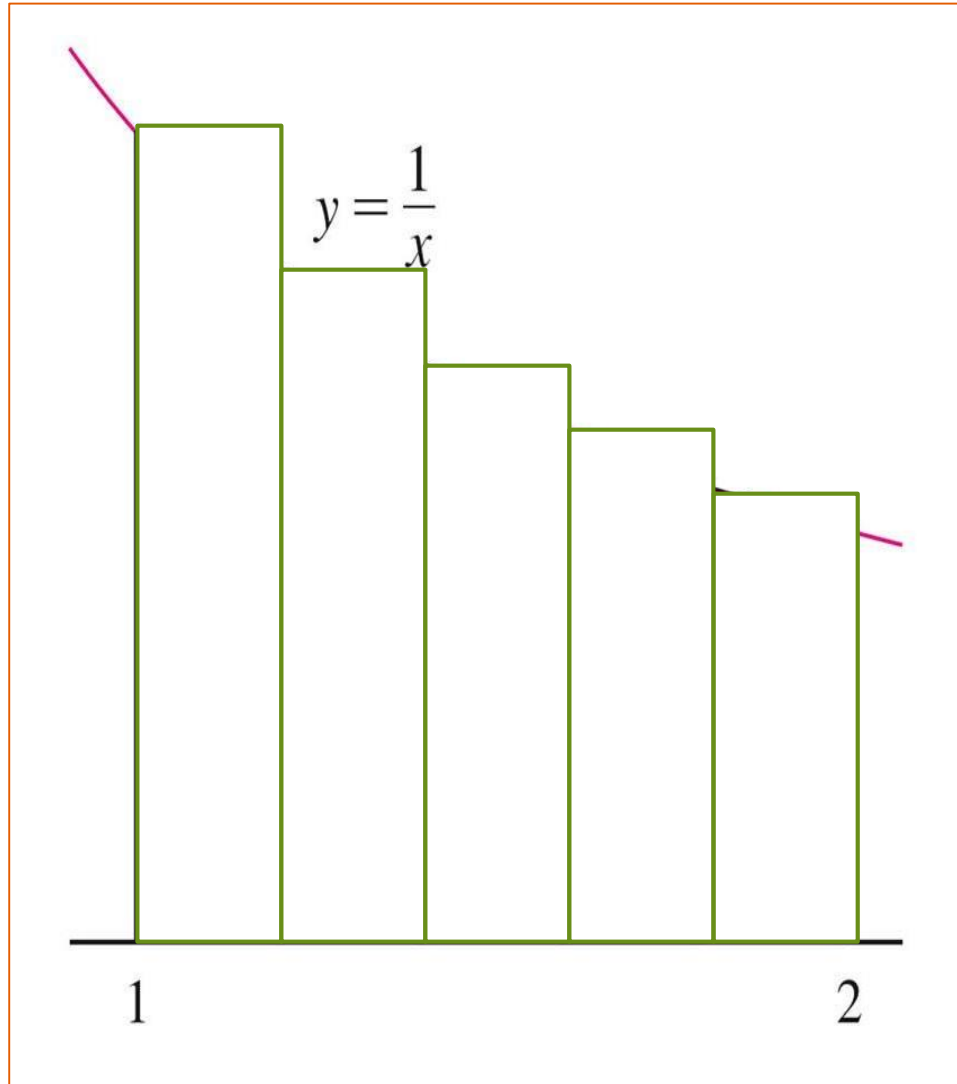
$$\int_1^2 \frac{1}{x} dx = \ln x \Big|_1^2 = \ln 2 = 0.693147\dots$$

- ▶ The error is $|(0.693147 - 0.745639)| * 100 / 0.693147 = 7.57\%$

APPROXIMATE INTEGRATION

Example 1(a)

The L_5 approximation is illustrated here.

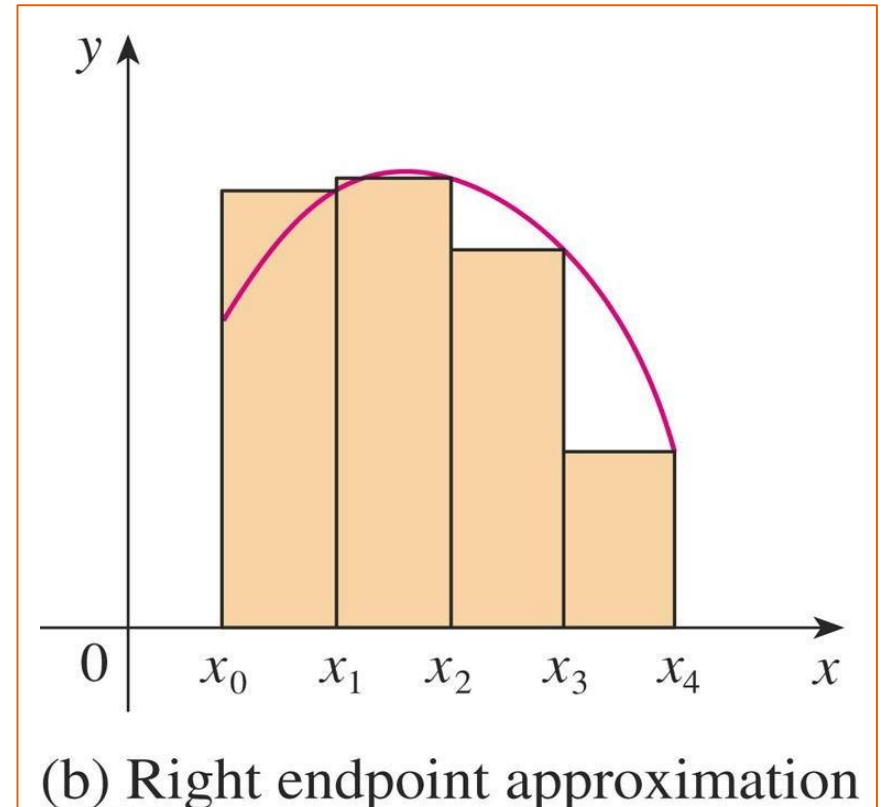


R_n APPROXIMATION

- If we choose x_i^* to be the right endpoint, $x_i^* = x_i$ and we have:

$$\int_a^b f(x) dx \approx R_n = \sum_{i=1}^n f(x_i) \Delta x \quad \text{Equation-2}$$

- The approximation R_n is called right endpoint approximation.



APPROXIMATE INTEGRATION

Example 1(a)

- ▶ Approximate the integral of $f(x) = 1/x$ with $n = 5$, using right end-point approximation: $\int_1^2 (1/x) dx$
- ▶ With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
- ▶ So, R_n the right endpoint approximation is :

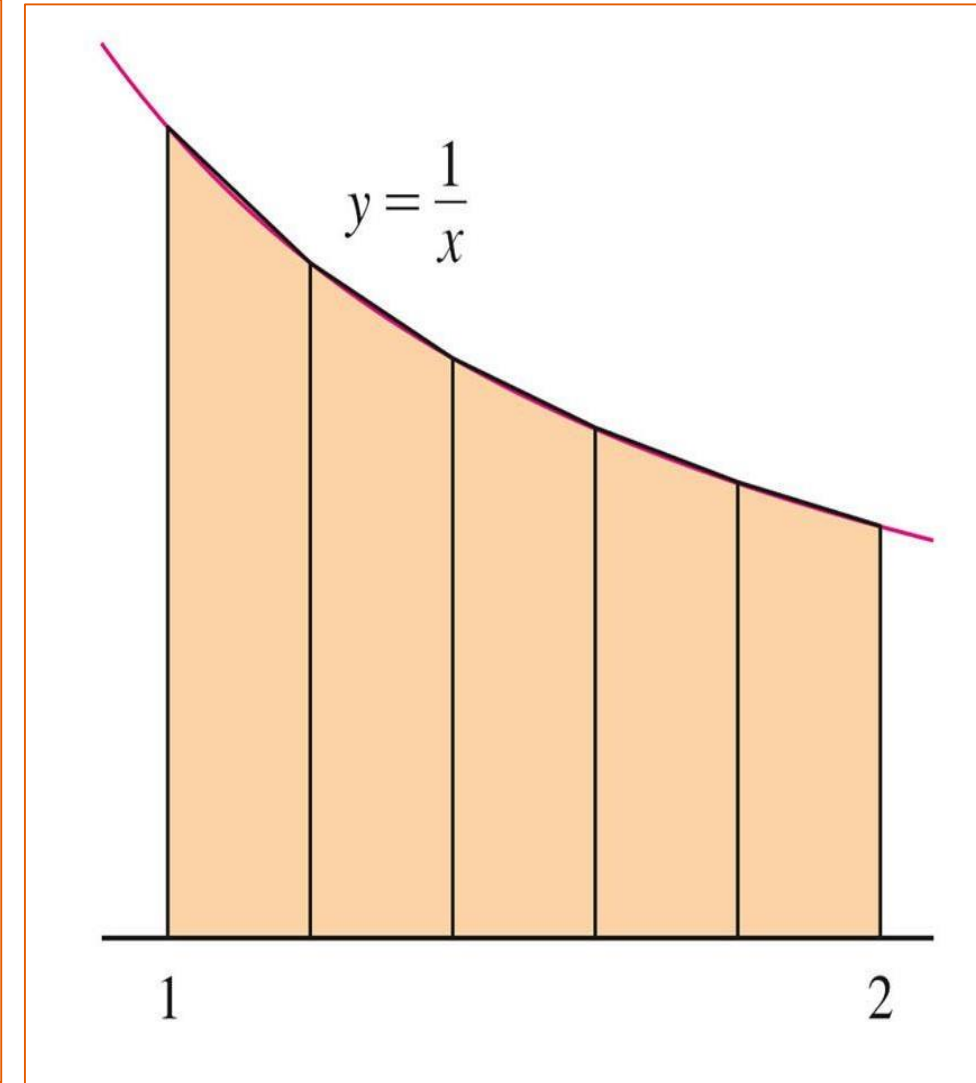
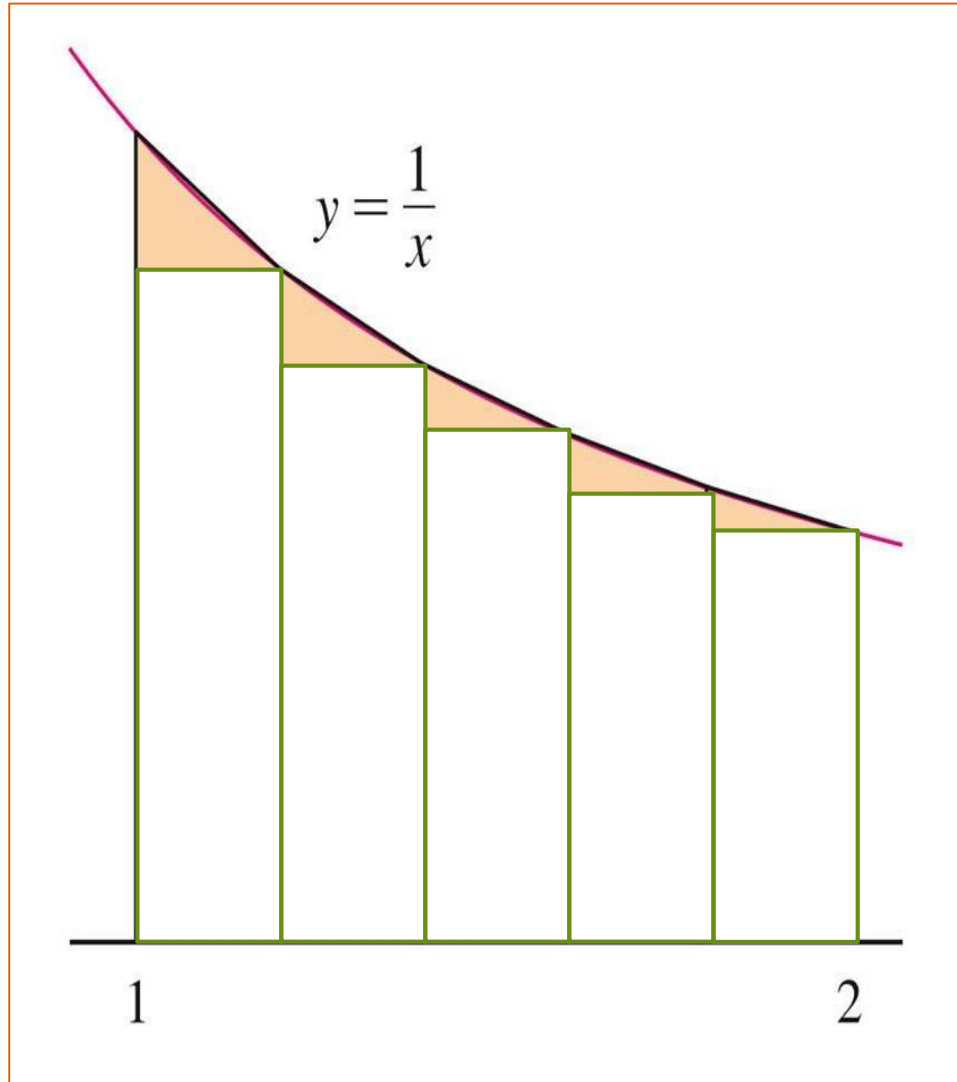
$$\int_1^2 \frac{1}{x} dx \approx R_5 = [f(1.2) + f(1.4) + f(1.6) + f(1.8) + f(2.0)] \times 0.2$$

$$= 0.2 \left(\frac{1}{1.2} + \frac{1}{1.4} + \frac{1}{1.6} + \frac{1}{1.8} + \frac{1}{2.0} \right) = 0.645634921(0.693147)$$

APPROXIMATE INTEGRATION

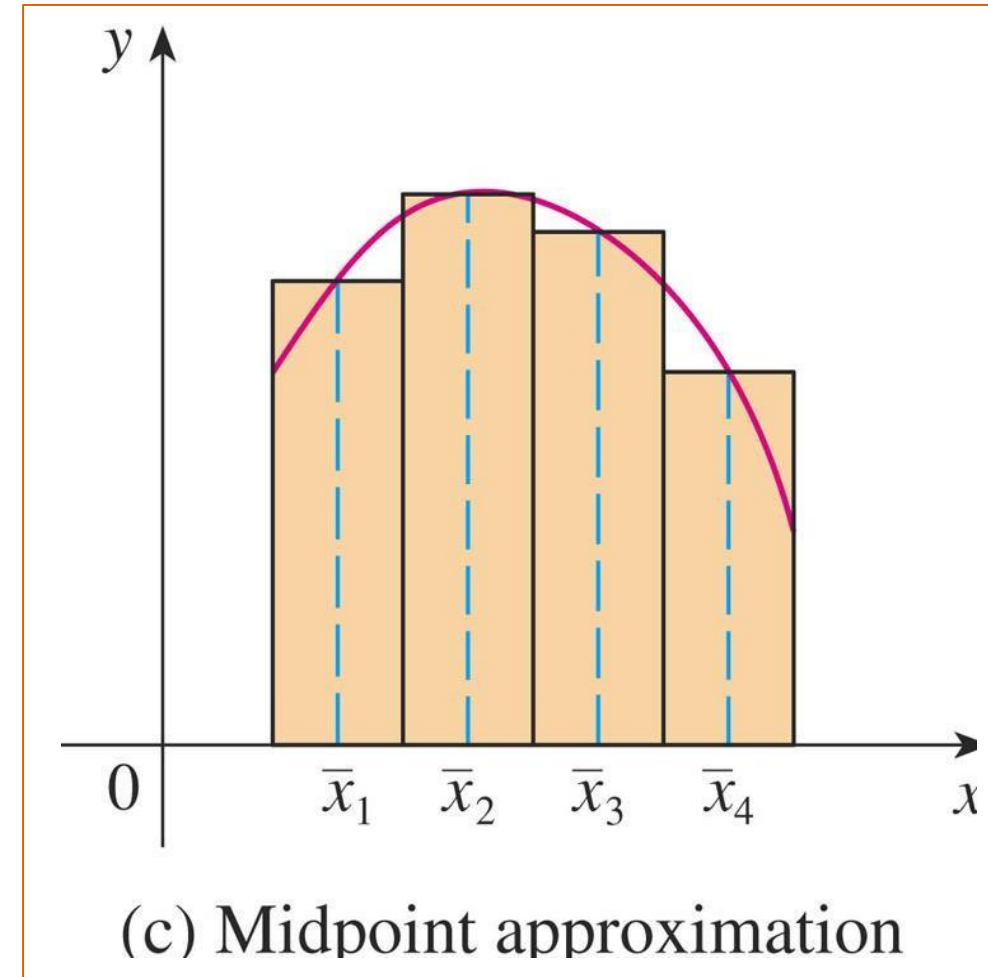
Example 1(a)

The R_5 approximation is illustrated here.



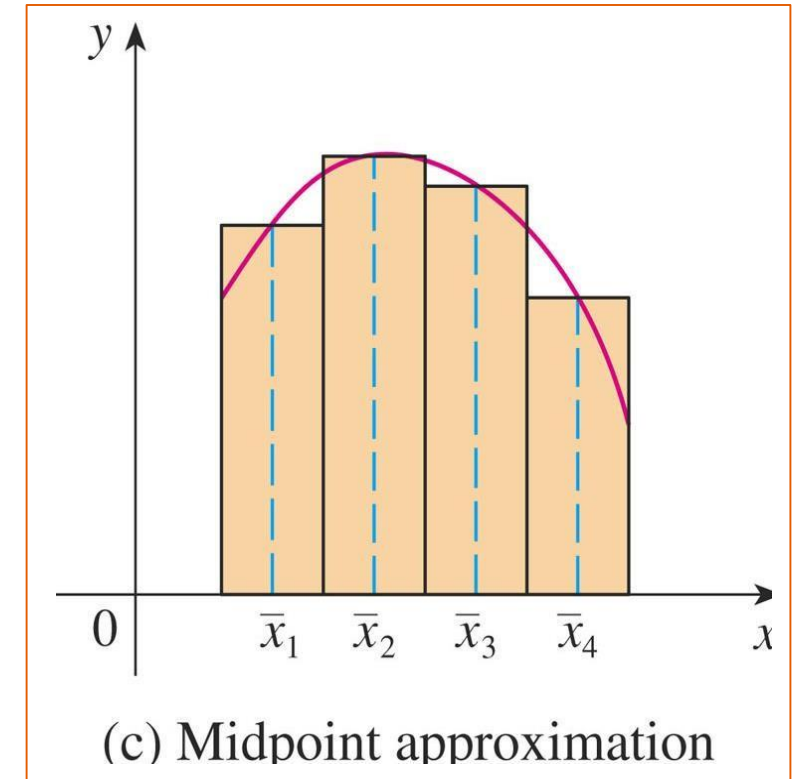
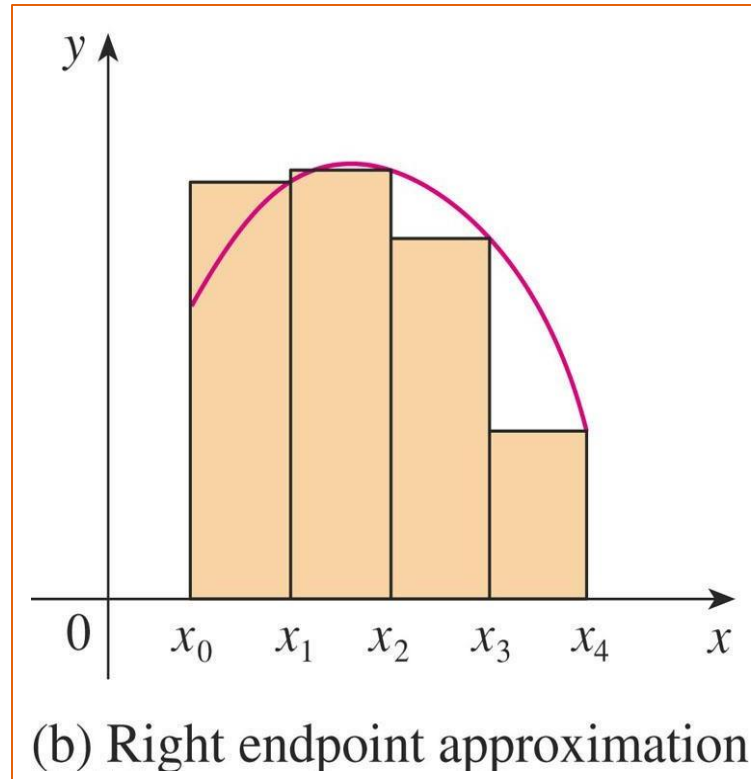
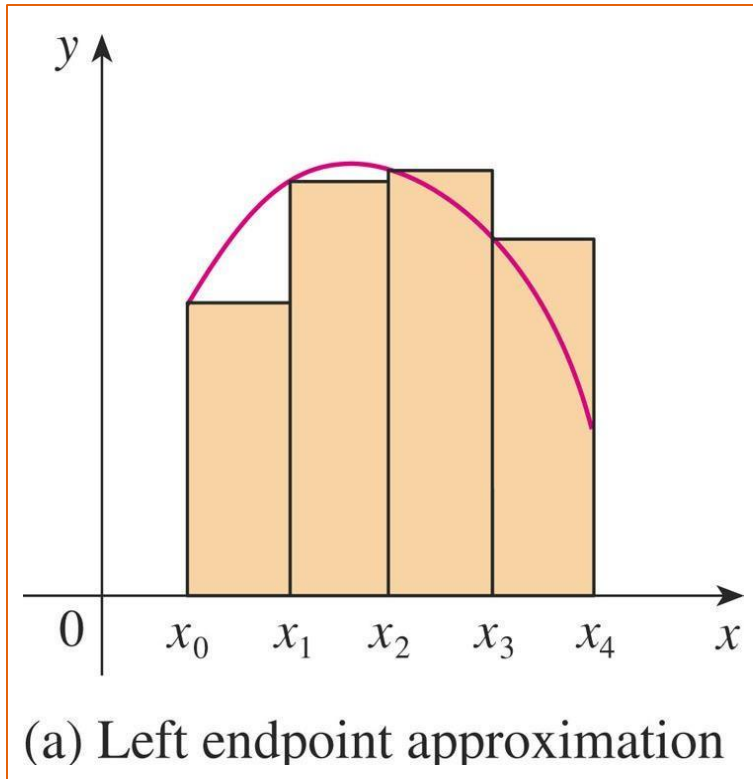
M_n APPROXIMATION

- ▶ We also considered the case where x_i^* is chosen to be the midpoint $\bar{x}_i = (x_{i-1} + x_i)/2$ of the subinterval $[x_{i-1}, x_i]$.
- ▶ The figure shows the midpoint approximation M_n .



M_n APPROXIMATION

- M_n appears to be better than either L_n or R_n .



THE MIDPOINT RULE

$$\int_a^b f(x) dx \approx M_n$$

$$= \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \dots + f(\bar{x}_n)]$$

► where $\Delta x = \frac{b-a}{n}$

► and $\bar{x}_i = \frac{1}{2} (x_{i-1} + x_i) = \text{midpoint of } [x_{i-1}, x_i]$

APPROXIMATE INTEGRATION

Example 1(a)

► Approximate the integral of $f(x) = 1/x$ with $n = 5$, using mid-point approximation: $\int_1^2 (1/x) dx$

► With $n = 5$, $a = 1$ and $b = 2$, we have:

$$\Delta x = (2 - 1)/5 = 0.2$$

► So, R_n the mid-point approximation is :

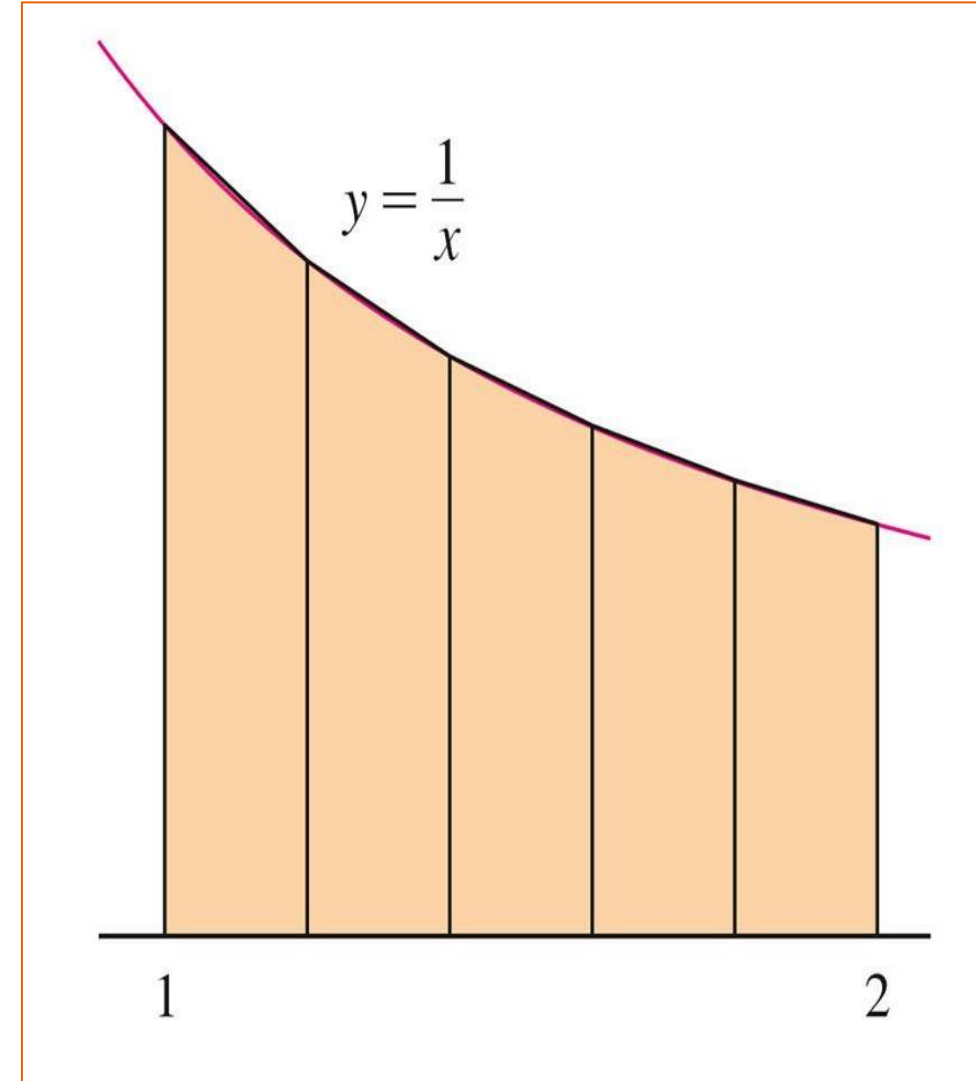
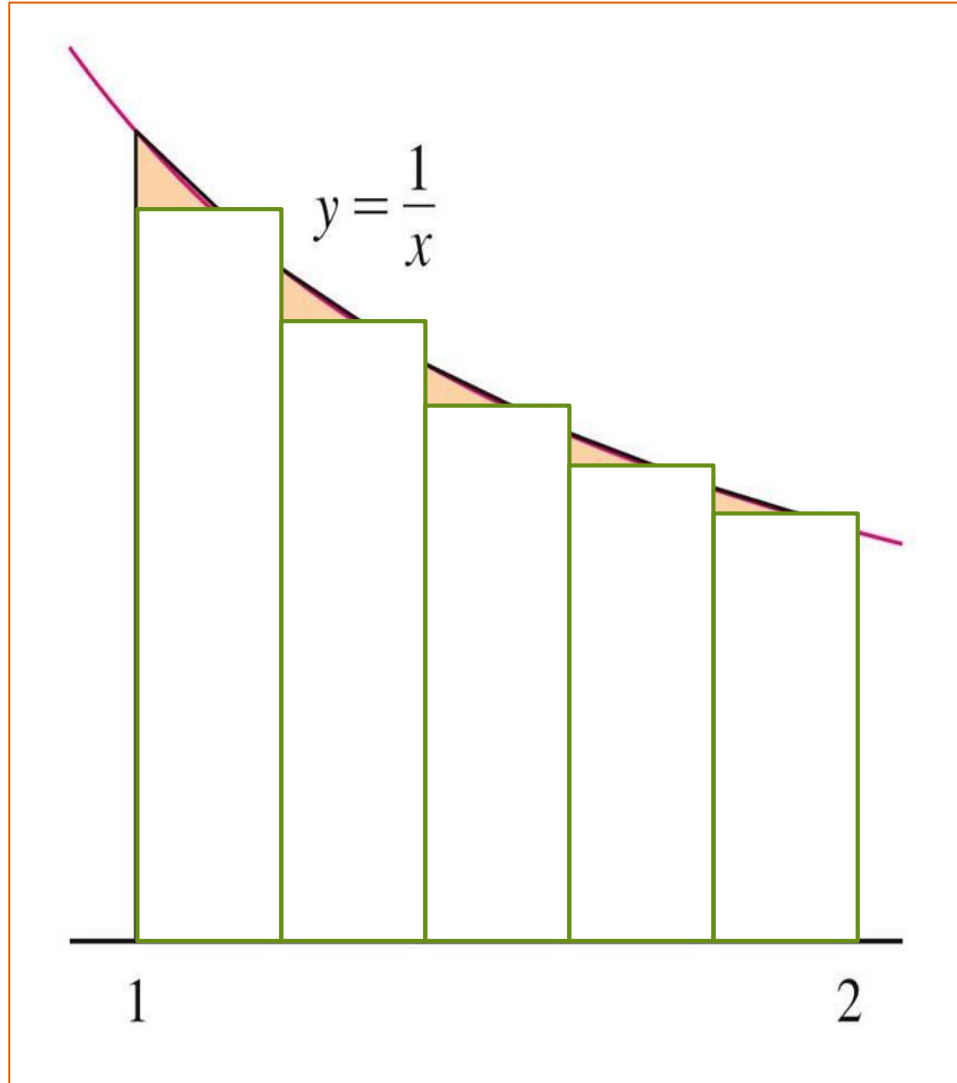
$$\int_1^2 \frac{1}{x} dx \approx R_5 = [f(1.1) + f(1.3) + f(1.5) + f(1.7) + f(1.9)] \times 0.2$$

$$= 0.2 \left(\frac{1}{1.1} + \frac{1}{1.3} + \frac{1}{1.5} + \frac{1}{1.7} + \frac{1}{1.9} \right) = 0.691907886 \text{ (0.693147)}$$

APPROXIMATE INTEGRATION

Example 1(a)

The M_5 approximation is illustrated here.



Why the Midpoint Rule is Better

Balances the Error: For a **monotonic (continuously increasing or decreasing)** function, the left endpoint **underestimates** the integral, and the right endpoint **overestimates** it. The midpoint essentially averages these two, partially canceling out the errors and yielding a much closer value to the exact area

- **Left End:** Uses the height at the left edge of each subinterval. It tends to underestimate the true area if the function is increasing, and overestimate it if it is decreasing.
- **Right End:** Uses the height at the right edge of each subinterval. It will generally overestimate for increasing functions and underestimate for decreasing ones.
- **Midpoint:** Uses the height exactly in the middle of each interval, which closely mirrors the "average height" of that specific section of the curve

TRAPEZOIDAL RULE

- ▶ Another approximation—called the Trapezoidal Rule—results from averaging the approximations in Equations 1 and 2, as follows.

$$\begin{aligned}\int_a^b f(x)dx &\approx \frac{1}{2} \left[\sum_{i=1}^n f(x_{i-1})\Delta x + \sum_{i=1}^n f(x_i)\Delta x \right] \\ &= \frac{\Delta x}{2} \left[\sum_{i=1}^n (f(x_{i-1}) + f(x_i)) \right] \\ &= \frac{\Delta x}{2} [(f(x_0) + f(x_1)) + (f(x_1) + f(x_2)) + \dots + (f(x_{n-1}) + f(x_n))] \\ &= \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)]\end{aligned}$$

TRAPEZOIDAL RULE

Approximate integration results from using straight line segments to approximate a curve.

$$\int_a^b f(x) dx \approx T_n$$

$$= \frac{\Delta x}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)]$$

► where $\Delta x = (b - a)/n$ and $x_i = a + i \Delta x$

Derivation of the Trapezoidal Rule

The area under the curve is a trapezoid. The integral

$$\begin{aligned} \int_a^b f(x)dx &\approx \text{Area of trapezoid} \\ &= \frac{1}{2}(\text{Sum of parallel sides})(\text{height}) \\ &= \frac{1}{2}(f(b) + f(a))(b - a) \\ &= (b - a) \left[\frac{f(a) + f(b)}{2} \right] \end{aligned}$$

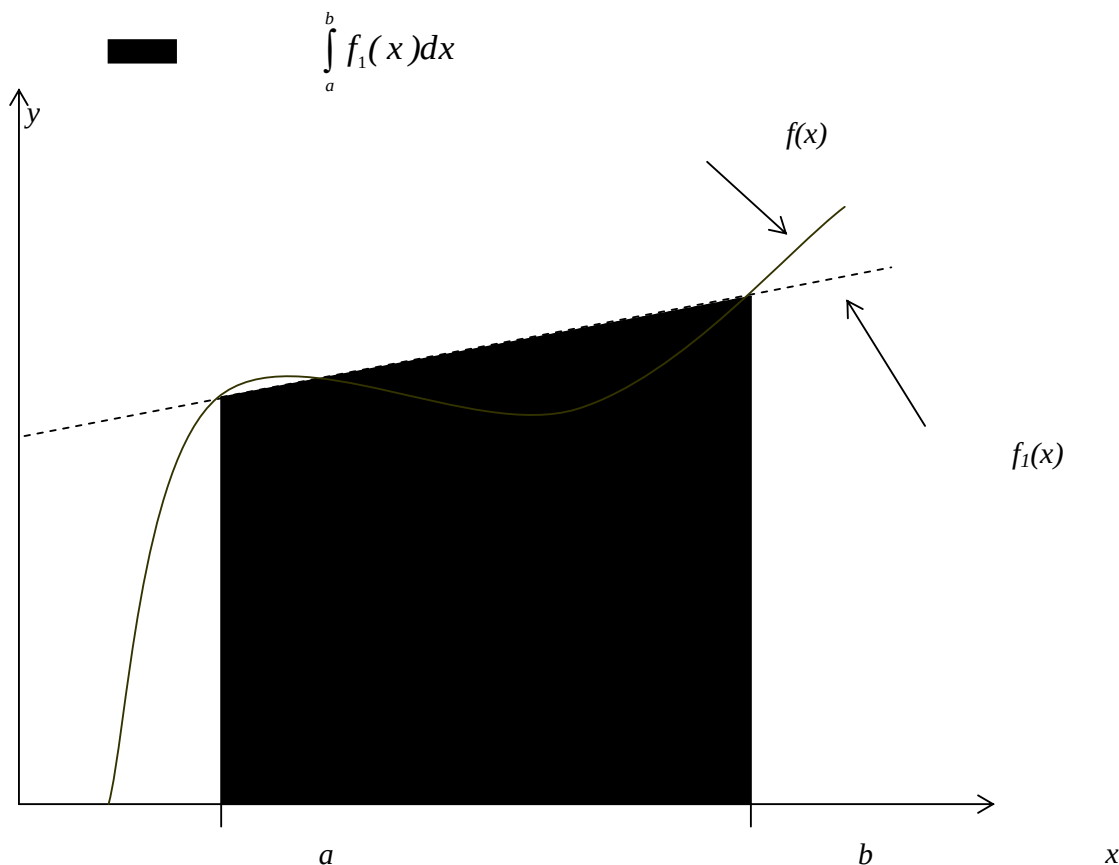


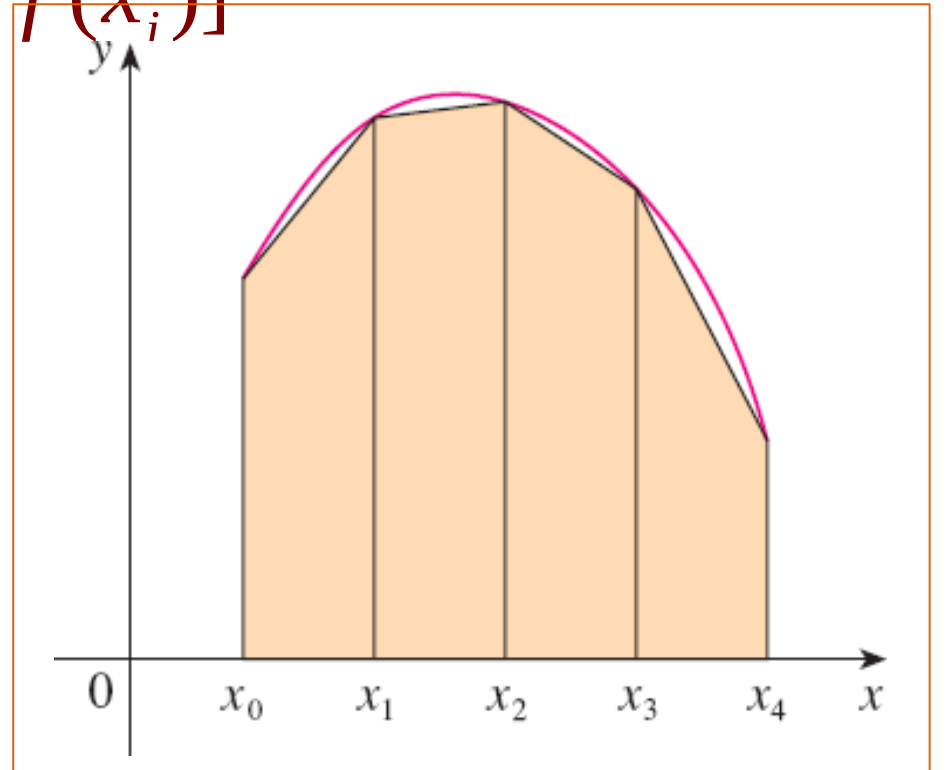
Figure 2: Geometric Representation

TRAPEZOIDAL RULE

- ▶ The reason for the name can be seen from the figure, which illustrates the case $f(x) \geq 0$.
- ▶ The area of the trapezoid that lies above the i_{th} subinterval is:

$$\Delta x \left(\frac{f(x_{i-1}) + f(x_i)}{2} \right) = \frac{\Delta x}{2} [f(x_{i-1}) + f(x_i)]$$

- ▶ If we add the areas of all these trapezoids, we get the right side of the Trapezoidal Rule.



APPROXIMATE INTEGRATION

Example 1(a)

- ▶ Approximate the integral with $n = 5$, using: $\int_1^2 (1/x) dx$
- ▶ With $n = 5$, $a = 1$ and $b = 2$, we have: $\Delta x = (2 - 1)/5 = 0.2$
- ▶ So, the Trapezoidal Rule gives:

$$\int_1^2 \frac{1}{x} dx \approx T_5 = \frac{0.2}{2} [f(1) + 2f(1.2) + 2f(1.4) + 2f(1.6) + 2f(1.8) + f(2)]$$

$$= 0.1 \left(\frac{1}{1} + \frac{2}{1.2} + \frac{2}{1.4} + \frac{2}{1.6} + \frac{2}{1.8} + \frac{1}{2} \right) \approx 0.695635$$

▶ $M_5 = 0.691908$

Example 4

The vertical distance covered by a rocket from $t=8$ to $t=30$ seconds is given by:

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

- a) Use single segment Trapezoidal rule to find the distance covered.
- b) Find the true error, E_t for part (a).
- c) Find the absolute relative true error, $|\epsilon_a|$ for part (a).

Solution

a)
$$I \approx (b - a) \left[\frac{f(a) + f(b)}{2} \right]$$

$$a = 8 \quad b = 30$$

$$f(t) = 2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t$$

$$\begin{aligned} f(8) &= 2000 \ln \left[\frac{140000}{140000 - 2100(8)} \right] - 9.8(8) \\ &= 177.27 \text{ m/s} \end{aligned}$$

$$f(30) = 2000 \ln \left[\frac{140000}{140000 - 2100(30)} \right] - 9.8(30) = 901.67 \text{ m/s}$$

$$\begin{aligned} I &= (30 - 8) \left[\frac{177.27 + 901.67}{2} \right] \\ &= 11868 \text{ m} \end{aligned}$$

b) The exact value of the above integral is

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt = 11061 \text{ m}$$

$$\begin{aligned} E_t &= \text{True Value} - \text{Approximate Value} \\ &= 11061 - 11868 \\ &= -807 \text{ m} \end{aligned}$$

c) The absolute relative true error, $|\epsilon_t|$, would be

$$|\epsilon_t| = \left| \frac{11061 - 11868}{11061} \right| \times 100 = 7.2959\%$$

Multiple Segment Trapezoidal Rule

In Example 4, the true error using single-segment trapezoidal rule was large. We can divide the interval $[8,30]$ into $[8,19]$ and $[19,30]$ intervals and apply the Trapezoidal rule over each segment.

$$f(t) = 2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t$$

$$\begin{aligned} \int_8^{30} f(t) dt &= \int_8^{19} f(t) dt + \int_{19}^{30} f(t) dt \\ &= (19-8) \left[\frac{f(8) + f(19)}{2} \right] + (30-19) \left[\frac{f(19) + f(30)}{2} \right] \end{aligned}$$

Multiple Segment Trapezoidal Rule

With

$$f(8) = 177.27 \text{ m/s}$$

$$f(30) = 901.67 \text{ m/s}$$

$$f(19) = 484.75 \text{ m/s}$$

The true error is:

$$E_t = 11061 - 11266$$

$$= -205 \text{ m}$$

Hence:

$$\int_8^{30} f(t) dt = (19 - 8) \left[\frac{177.27 + 484.75}{2} \right] + (30 - 19) \left[\frac{484.75 + 901.67}{2} \right] = 11266 \text{ m}$$

The true error now is reduced from **-807 m** to **-205 m**.

Extending this procedure to divide the interval into equal segments to apply the Trapezoidal rule, the sum of the results obtained for each segment is the approximate value of the integral.

Multiple Segment Trapezoidal Rule

Table 1 gives the values obtained using the multiple-segment Trapezoidal rule for

$$x = \int_8^{30} \left(2000 \ln \left[\frac{140000}{140000 - 2100t} \right] - 9.8t \right) dt$$

Exact Value=11061 m

n	Value	E_t	 ε_t %	 ε_a %
1	11868	-807	7.296	---
2	11266	-205	1.853	5.343
3	11153	-91.4	0.8265	1.019
4	11113	-51.5	0.4655	0.3594
5	11094	-33.0	0.2981	0.1669
6	11084	-22.9	0.2070	0.09082
7	11078	-16.8	0.1521	0.05482
8	11074	-12.9	0.1165	0.03560

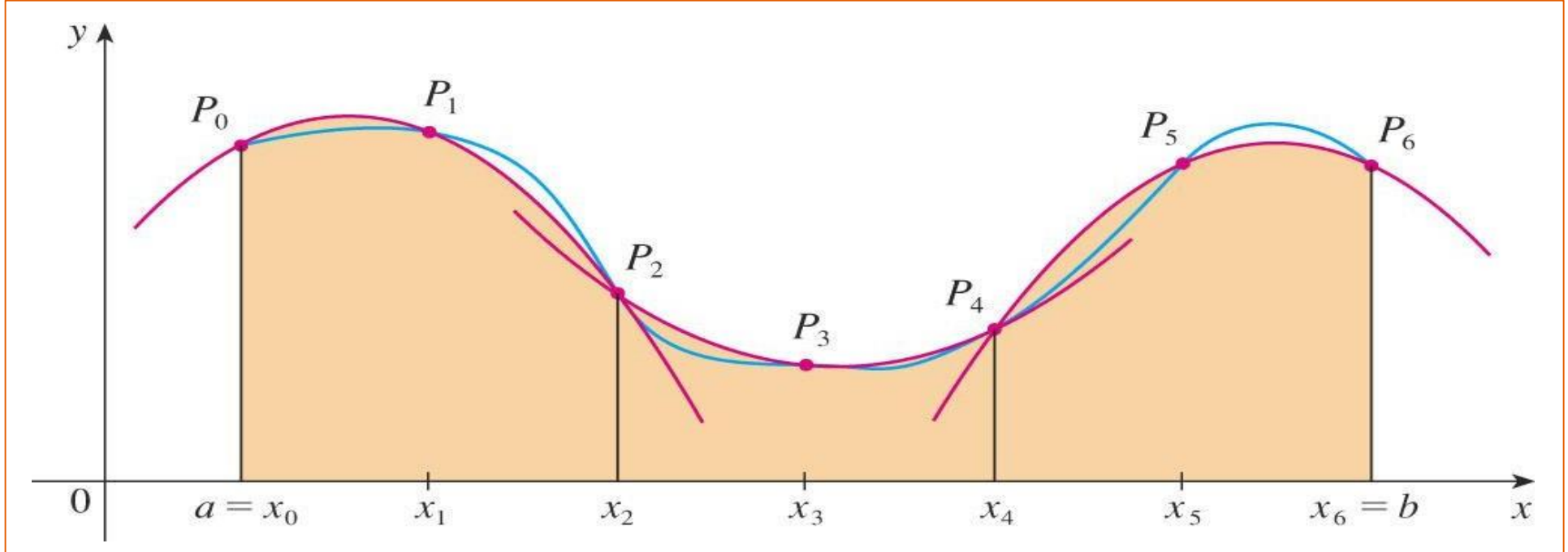
Table 1: Multiple Segment Trapezoidal Rule Values

APPROXIMATE INTEGRATION

- ▶ Another rule for approximate integration results from using (quadratic or cubic) curves instead of straight line segments to approximate a curve.
- ▶ As before, we divide $[a, b]$ into n subintervals of equal length $h = \Delta x = (b - a)/n$.
- ▶ However, this time, we assume n is an even number.

APPROXIMATE INTEGRATION

- ▶ Then, on each consecutive pair of intervals, we approximate the curve $y = f(x) \geq 0$ by a parabola, as shown.
- ▶ If $y_i = f(x_i)$, then $P_i(x_i, y_i)$ is the point on the curve lying above x_i .
- ▶ A typical parabola/curve passes through three consecutive points: P_{i-1} , P_i , P_{i+1}



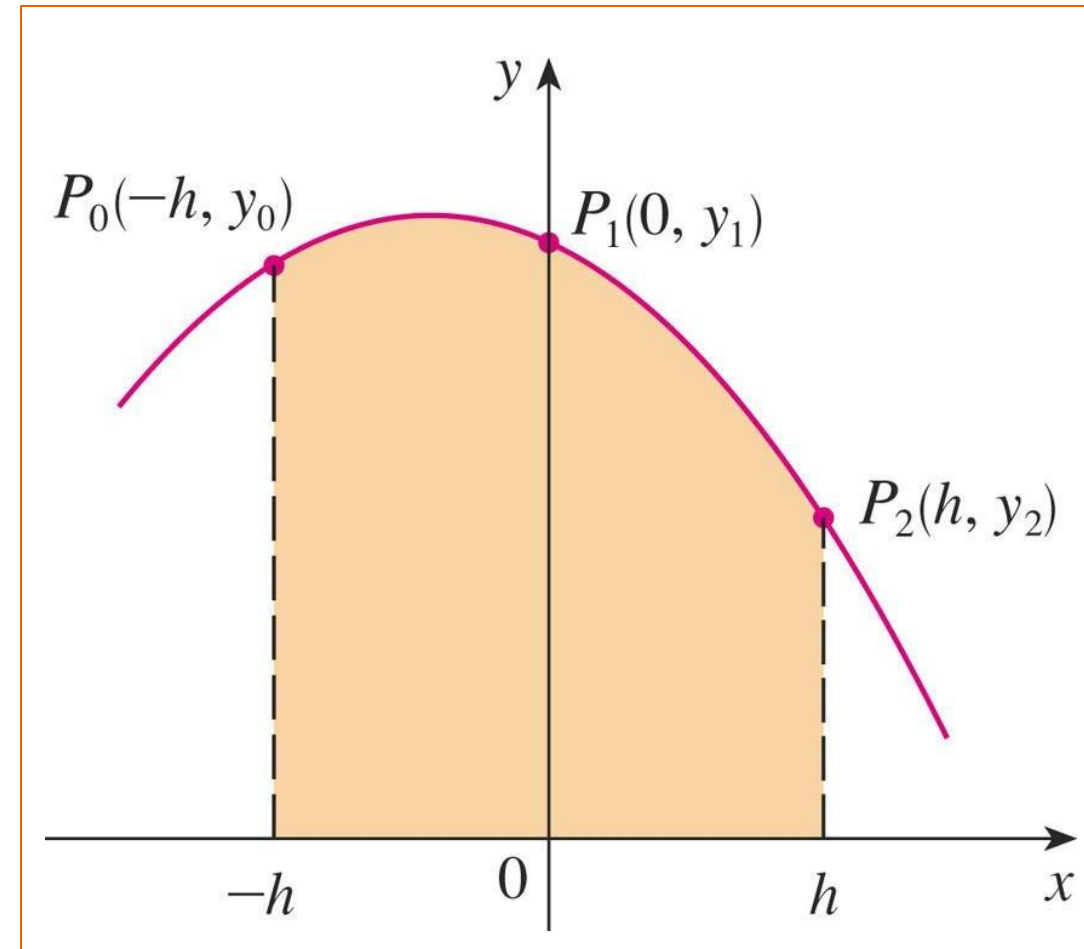
APPROXIMATE INTEGRATION

- To simplify our calculations, we first consider the case where:

$$x_0 = -h, x_1 = 0, x_2 = h$$

- We know that the equation of the parabola through P_0 , P_1 and P_2 is of the form

- $y = f(x) = Ax^2 + Bx + C$



APPROXIMATE INTEGRATION

Therefore, the area under the parabola from $x = -h$ to $x = h$ is:

$$\begin{aligned}\int_{-h}^h (Ax^2 + Bx + C)dx &= \left[A\frac{x^3}{3} + B\frac{x^2}{2} + Cx + K \right]_{-h}^h \\&= \left(A\frac{h^3}{3} + B\frac{h^2}{2} + Ch + K \right) - \left(-A\frac{h^3}{3} + B\frac{h^2}{2} - Ch + K \right) \\&= 2 \left(A\frac{h^3}{3} + Ch \right) \\&= \frac{h}{3} (2Ah^2 + 6C)\end{aligned}$$

APPROXIMATE INTEGRATION

However, as the parabola passes through $P_0(-h, y_0)$, $P_1(0, y_1)$, and $P_2(h, y_2)$, we have:

$$\begin{aligned} y_0 &= A(-h)^2 + B(-h) + C \\ &= Ah^2 - Bh + C \end{aligned}$$

$$y_1 = C \text{ or, } 4y_1 = 4C$$

$$y_2 = Ah^2 + Bh + C$$

APPROXIMATE INTEGRATION

► Therefore,

$$y_0 + 4y_1 + y_2 = 2Ah^2 + 6C$$

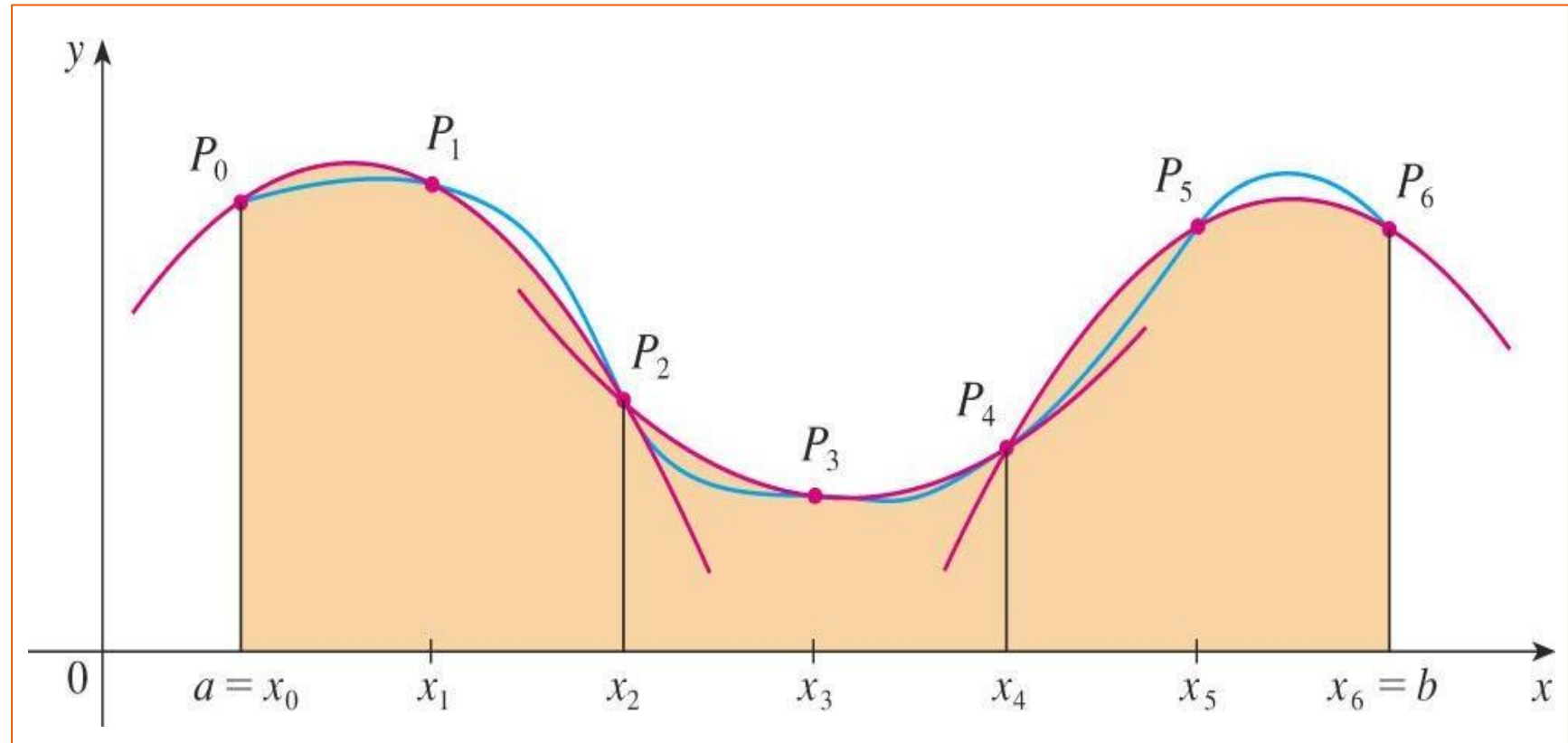
► So, we can rewrite the area under the parabola as:

$$\frac{h}{3} (y_0 + 4y_1 + y_2)$$

APPROXIMATE INTEGRATION

- ▶ Now, by shifting this parabola horizontally, we do not change the area under it.
- ▶ This means that the area under the parabola through P_0 , P_1 , and P_2 from $x = x_0$ to $x = x_2$ is still:

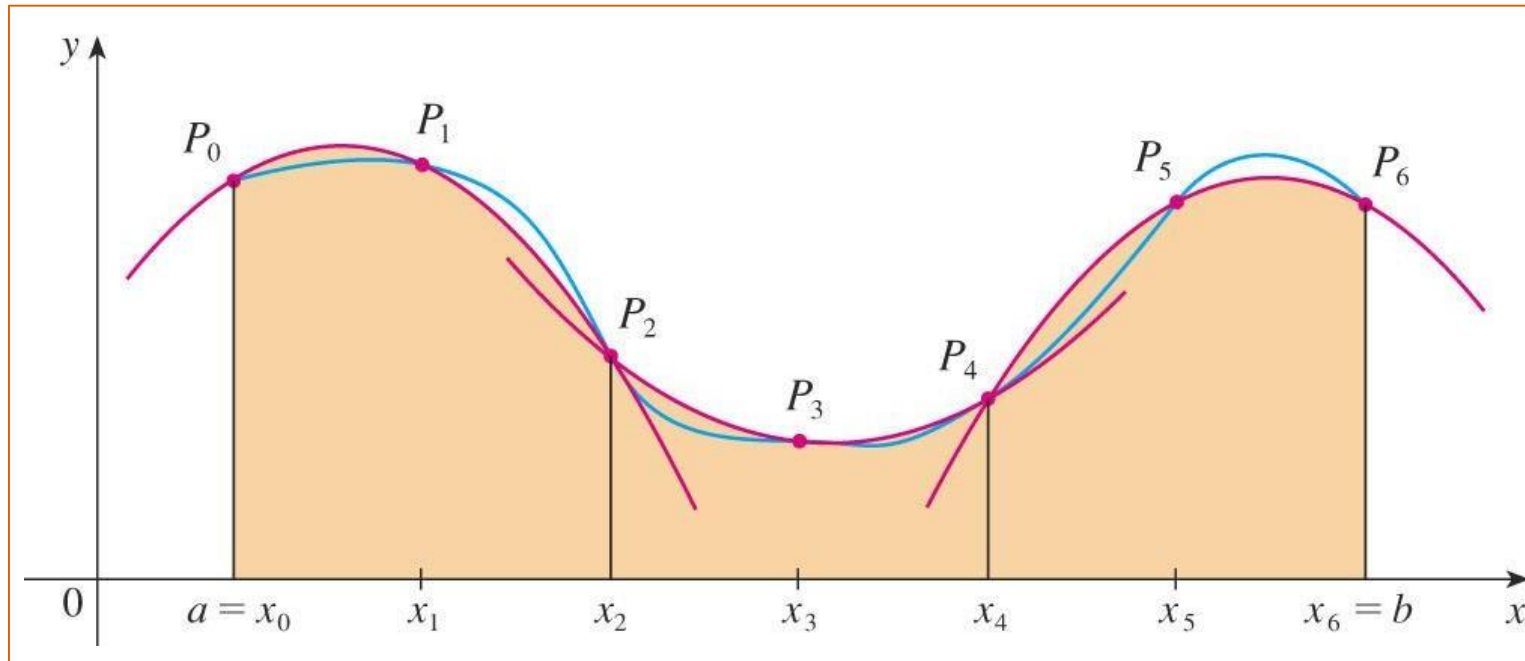
$$\frac{h}{3}(y_0 + 4y_1 + y_2)$$



APPROXIMATE INTEGRATION

- ▶ Similarly, the area under the parabola through P_2 , P_3 , and P_4 from $x = x_2$ to $x = x_4$ is:

$$\frac{h}{3}(y_2 + 4y_3 + y_4)$$



APPROXIMATE INTEGRATION

- Thus, if we compute the areas under all the parabolas and add the results, we get:

$$\int_a^b f(x)dx \approx \frac{h}{3}(y_0 + 4y_1 + y_2) + \frac{h}{3}(y_2 + 4y_3 + y_4) + \dots + \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n)$$

$$= \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + \dots + 2y_{n-2} + 4y_{n-1} + y_n)$$

- Note the pattern of coefficients: 1, 4, 2, 4, 2, 4, 2, . . . , 4, 2, 4, 1

SIMPSON'S 1/3 RULE

- This is called Simpson's 1/3 Rule—after the English the English mathematician Thomas Simpson (1710–1761).

$$\begin{aligned}\int_a^b f(x) dx &\approx S_n \\ &= \frac{\Delta x}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) \\ &\quad + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]\end{aligned}$$

- where n is even and h or $\Delta x = (b - a)/n$.

SIMPSON'S RULE

Example 5

- Use Simpson's 1/3 Rule with $n = 10$ to approximate

$$\int_1^2 (1/x) dx$$

- Putting $f(x) = 1/x$, $n = 10$, and $\Delta x = 0.1$ in Simpson's Rule, we obtain:

$$\begin{aligned}\int_1^2 \frac{1}{x} dx &\approx S_{10} \\ &= \frac{\Delta x}{3} [f(1) + 4f(1.1) + 2f(1.2) + 4f(1.3) + \dots + 2f(1.8) + 4f(1.9) + f(2)] \\ &= \frac{0.1}{3} \left(\frac{1}{1} + \frac{4}{1.1} + \frac{2}{1.2} + \frac{4}{1.3} + \frac{2}{1.4} + \frac{4}{1.5} + \frac{2}{1.6} + \frac{4}{1.7} + \frac{2}{1.8} + \frac{4}{1.9} + \frac{1}{2} \right)\end{aligned}$$

$$\approx 0.693150$$

SIMPSON'S RULE

- ▶ *In Example 5, notice that Simpson's Rule gives a much better approximation ($S_{10} \approx 0.693150$) to the true value of the integral ($\ln 2 \approx 0.693147$) than does either:*
- ▶ *Trapezoidal Rule ($T_{10} \approx 0.693771$)*
- ▶ *Midpoint Rule ($M_{10} \approx 0.692835$)*

SIMPSON's 3/8 RULE

- We know that the equation of the parabola through $P_0(x_0, y_0)$, $P_1(x_1, y_1)$, $P_2(x_2, y_2)$ and $P_3(x_3, y_3)$ is of the form

$$y = f(x) = Ax^3 + Bx^2 + Cx + D$$

or $y = a_0 + a_1x + a_2x^2 + a_3x^3$ (cubic polynomial)

- And the area under the parabola/curve of 4 points is

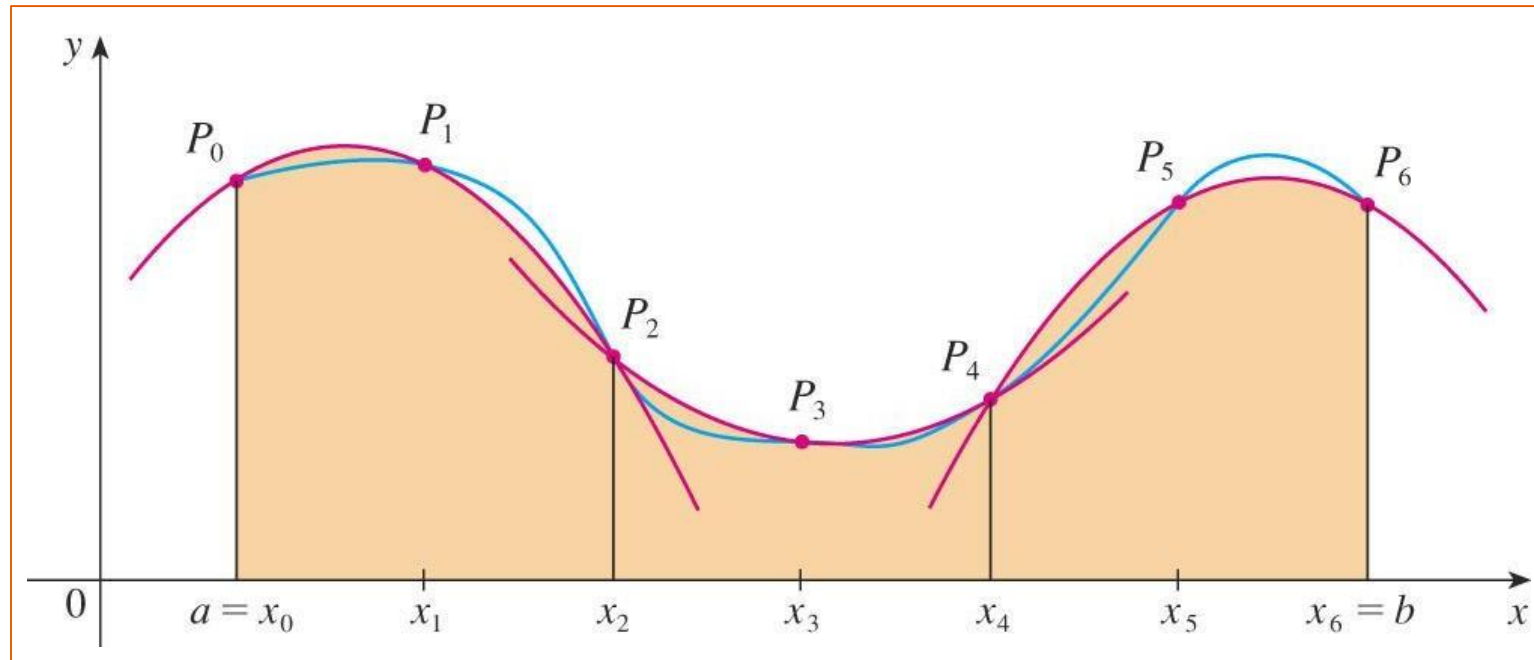
$$\frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3)$$

SIMPSON'S 3/8 RULE

- ▶ Thus, if we compute the areas under all the parabolas and add the results, we get:

$$\int_a^b f(x)dx \approx \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3) + \frac{3h}{8} (y_3 + 3y_4 + 3y_5 + y_6) + \dots + \frac{3h}{8} (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

- ▶ Note the pattern of coefficients: 1, 3, 3, 2, 3, 3, 2, ..., 2, 3, 3, 1



SIMPSON's 3/8 RULE

- ▶ Use Simpson's 3/8 Rule with $n = 6$ to approximate

$$\int_1^2 (1/x) dx$$

- ▶ Here $\Delta x = (b-a)/n = 1/6 = 0.167$
- ▶ We have the following tables of values.

x	1	1.166667	1.333333	1.5	1.666667	1.833333	2
f(x)	1	0.857143	0.75	0.666667	0.6	0.545455	0.5

SIMPSON's 3/8 RULE

- Now, we compute the value of the integral.

$$\int_1^2 \frac{1}{x} dx \approx \frac{3\Delta x}{8} [f(1.0) + f(2.0) + 2f(1.5) + 3\{f(1.167) + f(1.33) + f(1.667) + f(1.833)\}]$$

$$= \frac{3 \times 0.16667}{8} [1.0 + 0.5 + 2 \times 0.6667 + 3\{0.8571 + 0.75 + 0.6 + 0.5454\}]$$

$$= 0.06249 * 11.0911$$

$$= 0.6932$$